

STAT 332 Sampling and Experimental Design

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1 Sampling

Lecture 1, 2025/09/03

1.1 Intro & Survey Terminology

Assignments are there to help you prepare for exams!

Why study survey sampling?

- Obtain data for a question of interest from a population.
- Cannot take a census.
- Finite time and resources.

Definition 1.1 (Target Population).

The complete collection of individuals we wish to study and draw conclusions about. Denoted by $U = \{1, 2, \dots, N\}$.

Definition 1.2 (Sampled (Study) Population).

The collection of individuals that could be included by our sampling design.

Definition 1.3 (Sampling Unit).

The object being sampled.

Definition 1.4 (Frame).

The list of all sampling units.

Definition 1.5 (Observation Unit).

The object on which we take measurements.

Note. Often, this is the person we ask questions to.

Three main steps at which error and bias can be introduced in the conducting of a survey:

- Constructing the frame (coverage error or study error)

- Obtaining the sample/respondents (sampling bias/error, non-response error)
- Measurement of the response (measurement error)

Definition 1.6 (Sampling Design (Sampling Protocol)).

A **sampling design** (or sampling protocol) is the mechanism by which we select our samples.

Note. **Sampling error** is the estimation error due to sampling variability.

Definition 1.7 (Selection Bias).

Differences between the target population and the effective sampled population.

Note. Ideally, target population = population that can be effectively sampled.

Sources of selection bias:

- Coverage: e.g., can be difficult to construct a complete frame
- Sampling: e.g., improper sampling design
- Non-response: people don't respond or are not reachable

Definition 1.8 (Measurement Error).

The measured values of the attributes on the observation units are different from the true values.

Lecture 2, 2025/09/08

1.2 Probability-Based Sampling Design

Assume that frame is complete (ideal case). Assume also that $U = \{1, 2, \dots, N\}$. We denote the responses (corresponding values for attribute of interest) by $y_1, y_2, \dots, y_N$.

Population quantities:

- $\mu = \frac{1}{N} \sum_{i \in U} y_i$ (population mean)
- $\tau = \sum_{i \in U} y_i$ (population total)
- $\sigma^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 = \frac{1}{N-1} [\sum_{i \in U} y_i^2 - N\mu^2]$ (population variance b/c population is finite).

Note.

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 \\
&= \frac{1}{N-1} \sum_{i \in U} (y_i^2 - 2\mu y_i + \mu^2) \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - 2\mu \sum_{i \in U} y_i + N\mu^2 \right] \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - 2N\mu^2 + N\mu^2 \right] \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - N\mu^2 \right].
\end{aligned}$$

Suppose we have a sample $s \subseteq U$, a subset of frame with sample size n .

Estimates of population quantities based on sample:

- $\hat{\mu} = \frac{1}{n} \sum_{i \in s} y_i$, sample mean estimates population mean
- $\hat{\tau} = \sum_{i \in s} y_i$, sample total estimates population total
- $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i \in s} (y_i - \hat{\mu})^2$, sample variance estimates population variance

Note. As in STAT 231, we treat estimates as constant values based on a specific sample, i.e. y_i 's are constants. We treat the sampling design/process as random, so s is a realization of a random variable S .

Sampling design tells us $\mathbb{P}(S = s)$ which captures the **inclusion probabilities**.

Definition 1.9 (Inclusion Probabilities).

$$\pi_i = \mathbb{P}(i \in S), \quad \pi_{ij} = \mathbb{P}(i, j \in S), \quad i \neq j \in U.$$

Then, we can define estimators as functions of S :

- $\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i$ (sample mean estimator)
- $\tilde{\tau} = \sum_{i \in S} y_i$ (sample total estimator)
- $\tilde{\sigma}^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \tilde{\mu})^2$ (sample variance estimator)

Definition 1.10 (Simple Random Sampling (SRS)).

SRS is also known as **SRS without replacement (SRSWOR)**. Take a sample of size n out of N , such that all possible samples of size n are equally likely.

Note. The randomness in SRS comes from the sampling design, not the responses.

Example. Let $N = 4$, $n = 2$, then there are $\binom{4}{2} = 6$ possible samples:

$$\{1, 2\} \quad \{1, 3\} \quad \{1, 4\} \quad \{2, 3\} \quad \{2, 4\} \quad \{3, 4\},$$

each with probability $\frac{1}{6}$, e.g. $\mathbb{P}(S = \{1, 2\}) = \frac{1}{6}$.

Question: How many possible samples of size n ?

Answer: $\binom{N}{n} = \frac{N!}{n!(N-n)!}$.

Thus, we have

$$\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}} \quad \text{for any subset } s \text{ of size } n.$$

We can also compute inclusion probabilities:

$$\pi_i = \mathbb{P}(i \in S) = \frac{\# \text{ of samples with unit } i \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{1}{1}\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{n}{N}.$$

Note. $\binom{1}{1}$ is the number of ways to choose unit i (to include i , we must pick i , so 1 way), and $\binom{N-1}{n-1}$ is the number of ways to choose the remaining $n - 1$ units from the other $N - 1$ units.

$$\pi_{ij} = \mathbb{P}(i, j \in S) = \frac{\# \text{ of samples with units } i, j \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{2}{2}\binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}.$$

Note. $\pi_{ij} \neq \pi_i\pi_j$ because sampling from a finite population without replacement means that the events $i \in S$ and $j \in S$ are not independent.

R Intro & SRS in R

Here is some basic syntax for R:

```
x <- c()      # creates an empty vector
for (i in 1:50) {
  x[i] <- 2*i    # fill it with some stuff
}
x             # look at what it now has
mean(x)
var(x)        # sample mean and variance
```

Obtaining SRS in R:

```
> N <- 10
> n <- 5
> x = c(1:10)
> sample(x,5, replace=F)    # replace=F indicates without replacement
[1] 6 2 4 7 8
> sample(x,5, replace=F)
[1] 8 4 5 6 3
```

Probability Review

Let X be a discrete random variable. Then,

$$(1) \mathbb{E}[X] = \sum_x x \mathbb{P}(X = x), \quad \mathbb{E}[g(X)] = \sum_x g(x) \mathbb{P}(X = x).$$

$$(2) \text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

$$(3) \text{Linearity: } \mathbb{E}\left[\sum_i^n a_i X_i\right] = \sum_i^n a_i \mathbb{E}[X_i].$$

$$(4) \text{Var}\left(\sum_i^n a_i X_i\right) = \sum_i^n a_i^2 \text{Var}(X_i) + 2 \sum_i^n \sum_{j \neq i}^n a_i a_j \text{Cov}(X_i, X_j).$$

$$(5) \text{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j].$$

$$(6) \text{ If } X_i \text{ and } X_j \text{ are independent, then } \text{Cov}(X_i, X_j) = 0.$$

Example. Let's define an indicator random variable. Let

$$I_i = \begin{cases} 1, & \text{if unit } i \text{ is selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i = 1) = \mathbb{P}(i \in S) = \pi_i$ and $\mathbb{P}(I_i = 0) = 1 - \pi_i$. Thus,

$$\mathbb{E}[I_i] = 0 \cdot (1 - \pi_i) + 1 \cdot \pi_i = \pi_i,$$

$$\mathbb{E}[I_i^2] = 0^2 \cdot (1 - \pi_i) + 1^2 \cdot \pi_i = \pi_i,$$

$$\text{Var}(I_i) = \mathbb{E}[I_i^2] - (\mathbb{E}[I_i])^2 = \pi_i - \pi_i^2 = \pi_i(1 - \pi_i).$$

For $i \neq j$, consider

$$I_i I_j = \begin{cases} 1, & \text{if both } i \text{ and } j \text{ are selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i I_j = 1) = \mathbb{P}(i, j \in S) = \pi_{ij}$ and $\mathbb{P}(I_i I_j = 0) = 1 - \pi_{ij}$. Thus,

$$\mathbb{E}[I_i I_j] = 0 \cdot (1 - \pi_{ij}) + 1 \cdot \pi_{ij} = \pi_{ij},$$

$$\text{Cov}(I_i, I_j) = \mathbb{E}[I_i I_j] - \mathbb{E}[I_i] \mathbb{E}[I_j] = \pi_{ij} - \pi_i \pi_j.$$

So, we can write $\tilde{\mu}$ in terms of indicators.

$$\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i = \frac{1}{n} \sum_{i=1}^N I_i y_i = \frac{1}{n} \sum_{i \in U} I_i y_i = \frac{1}{n} \sum_{i=1}^N I_i y_i.$$

Definition 1.11 (Bias, Unbiased Estimator).

Suppose that $\tilde{\theta}$ is an estimator for θ . Then, the **bias** of $\tilde{\theta}$ is

$$\text{Bias}(\tilde{\theta}) = \mathbb{E}[\tilde{\theta}] - \theta.$$

$\tilde{\theta}$ is an **unbiased estimator** for θ if $\text{Bias}(\tilde{\theta}) = 0$, i.e. $\mathbb{E}[\tilde{\theta}] = \theta$.

Question: What are $\mathbb{E}[\tilde{\mu}]$ and $\text{Var}(\tilde{\mu})$ for SRS?

Answer:

$$\begin{aligned}\mathbb{E}[\tilde{\mu}] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^N I_i y_i\right] = \frac{1}{n} \sum_{i=1}^N y_i \underbrace{\mathbb{E}[I_i]}_{\pi_i} \\ &= \frac{1}{n} \sum_{i=1}^N y_i \frac{n}{N} = \frac{1}{N} \sum_{i=1}^N y_i = \mu.\end{aligned}$$

So $\tilde{\mu}$ is an unbiased estimator for μ .

For SRS,

$$\text{Cov}(I_i, I_j) = \pi_{ij} - \pi_i \pi_j = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Then,

$$\begin{aligned}\text{Var}(\tilde{\mu}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^N I_i y_i\right) \\ &= \frac{1}{n^2} \left[\sum_{i=1}^N y_i^2 \underbrace{\text{Var}(I_i)}_{\pi_i(1-\pi_i)} + 2 \sum_{i=1}^N \sum_{j \neq i} y_i y_j \text{Cov}(I_i, I_j) \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \left[\sum_{i=1}^N y_i^2 - \frac{1}{N-1} \sum_{i=1}^N \sum_{j \neq i} y_i y_j \right]\end{aligned}$$

Note that $\sum_{i=1}^N \sum_{j \neq i} y_i y_j = \sum_{i=1}^N y_i \sum_{j \neq i} y_j = \sum_{i=1}^N y_i (N\mu - y_i) = N\mu \sum_{i=1}^N y_i - \sum_{i=1}^N y_i^2 = N^2\mu^2 - \sum_{i=1}^N y_i^2.$

$$\begin{aligned}&= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \left[\sum_{i=1}^N y_i^2 - \frac{1}{N-1} (N^2\mu^2 - \sum_{i=1}^N y_i^2) \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \frac{1}{N-1} \left[((N-1) + 1) \sum_{i=1}^N y_i^2 - N^2\mu^2 \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} N \frac{1}{N-1} \left(\sum_{i=1}^N y_i^2 - N\mu^2 \right) \\ &= \frac{1}{n} \left(1 - \frac{n}{N}\right) \sigma^2.\end{aligned}$$

Thus for SRS, $\mathbb{E}[\tilde{\mu}] = \mu$ and $\text{Var}(\tilde{\mu}) = \frac{1}{n} \left(1 - \frac{n}{N}\right) \sigma^2$.

To get CIs, we estimate $\text{Var}(\tilde{\mu})$ with $\widehat{\text{Var}}(\tilde{\mu}) = \frac{1}{n} \left(1 - \frac{n}{N}\right) \hat{\sigma}^2$, as it turns out that $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.